
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Multimodal Large Language Models (MLLMs) incur substantial computational overhead on long videos because effective video understanding requires large sequences of visual tokens. Existing efficiency methods reduce this burden mainly by pruning or merging tokens according to importance or similarity. Although these strategies remove redundancy, they underrepresent a defining property of video: semantic trajectories are often determined by changes, transitions, and turning points. They also typically lack a unified representation of spatial and temporal token relationships. We present ST-SimDiff, a training-free framework built on the principle that similarity should expose redundancy, whereas difference should preserve events. ST-SimDiff first constructs a spatio-temporal graph over visual tokens, thereby encoding local spatial adjacency and temporal continuity within a single structure. It then performs parallel dual selection: 1) similarity-based representative token selection applies community detection to retain central tokens from redundant and stable regions; and 2) temporal difference-based event token selection identifies content-changing points and preserves tokens associated with dynamic transitions. The resulting subset compactly represents persistent visual content together with event-defining changes. Extensive experiments show that ST-SimDiff consistently surpasses state-of-the-art token compression approaches while substantially reducing inference cost. Our code will be released upon acceptance.

1 INTRODUCTION

The rapid progress of Large Language Models (LLMs) has accelerated the development of Large Vision-Language Models (LVLMs), which now provide strong image and video understanding capabilities (Zhang et al., 2024c; Liu et al., 2024c). In video applications, contemporary LVLMs typically sample frames from a video and convert each frame into hundreds or thousands of visual tokens before forwarding the resulting sequence to the LLM (Bai et al., 2025; Li et al., 2024). This representation is expressive, but its token count grows quickly with video duration and resolution, producing substantial computational and memory costs. Efficient video-token processing is therefore a central requirement for long-video analysis and real-time multimodal interaction (Fu et al., 2024b).

Existing LVLM efficiency methods mainly reduce tokens through importance or similarity. Importance-based pruning discards visual tokens that receive low attention in later model layers, leveraging the redundancy observed in LVLM attention distributions over visual inputs (Chen et al., 2024). Similarity-based merging or selection instead exploits high feature similarity among visual tokens, either across adjacent frames or within individual frames. FrameFusion (Fu et al., 2024b), for example, reduces temporal redundancy by merging similar tokens from neighboring frames, whereas VisionZip (Yang et al., 2025) selects dominant tokens directly at the vision-encoder level.

Despite their effectiveness in reducing visual-token counts, existing methods leave two limitations that directly affect long-video understanding. First, many approaches model either spatial correlations within a frame or temporal correlations at corresponding positions across frames, rather than representing joint spatio-temporal structure. This separation limits their ability to exploit redun-

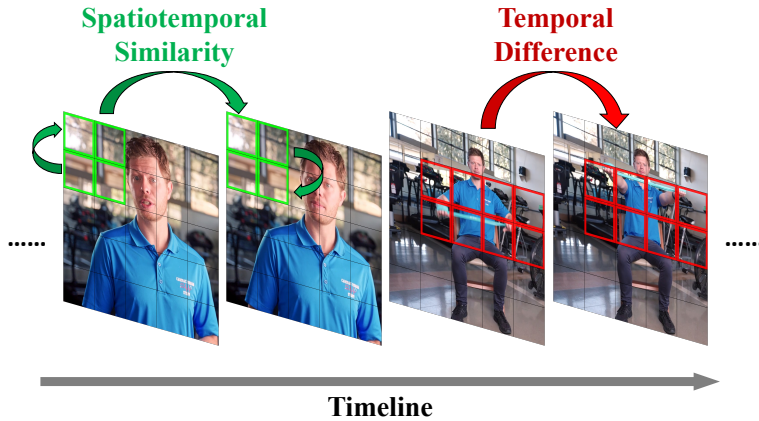


Figure 1: Motivation of ST-SimDiff. Efficient video understanding requires a joint account of spatiotemporal similarity and temporal difference: similarity (left) supports compression of redundant information within and across frames, whereas difference (right) identifies actions and events that determine the video trajectory.

dancy patterns produced by persistent objects, backgrounds, and motions. Second, most methods optimize for informational commonality, such as importance or similarity, but do not explicitly protect visual difference. Because video semantics are often determined by transitions, sudden actions, and event boundaries, a compression rule that favors only common or representative content may remove the tokens that explain what happened.

This paper advances a complementary view: similarity identifies redundancy, whereas difference identifies key moments. Under this view, effective video-token compression must satisfy two coupled objectives: it should represent stable visual content with few tokens while preserving the temporal changes that define events. We instantiate this principle as **ST-SimDiff**, a dual token-selection framework built on a spatio-temporal graph. In the first component, **similarity-based representative token selection**, all visual tokens are represented as graph nodes connected according to spatio-temporal feature similarity. Community detection identifies tightly connected token clusters corresponding to stable or persistent visual elements, and high-centrality tokens from each cluster are retained as representatives. In the second component, **difference-based event token selection**, temporal graph edges are examined to detect sharp drops in similarity between adjacent frames. These drops mark turning points, and the corresponding tokens are retained as event-critical evidence. The final visual input to the LLM is the union of representative tokens and event tokens.

Our contributions are as follows:

- We formulate video-token compression around the complementary roles of similarity and difference, establishing redundancy removal and event preservation as coupled objectives for efficient VLM inference.
- We introduce a spatio-temporal graph framework that models spatial and temporal token relationships in a unified structure, together with a dual selection strategy that retains representative stable tokens and pivotal event tokens.
- We provide extensive benchmark evidence showing that ST-SimDiff substantially reduces the visual-token budget while maintaining or improving video understanding performance across the evaluated settings.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has substantially accelerated multimodal understanding and has produced

a broad family of Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These models typically process images or video frames by converting them into visual tokens with pre-trained visual encoders and then passing those tokens to LLMs. Alignment modules, including MLPs (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023), commonly bridge the visual encoder and the LLM. This architecture supports strong multimodal performance (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a); in video understanding, it is especially important for longer, higher-resolution, and more complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

The same architecture also creates a pronounced efficiency bottleneck. Because video models process continuous frame sequences, their visual-token sequences are far larger than those produced by static-image inputs, increasing both training and inference cost. Efficient processing of large video-token sequences, and especially the reduction of unnecessary visual tokens, therefore remains a central problem in LVLM research.

2.2 VISUAL TOKEN COMPRESSION

Driven by the overhead of video processing, visual-token compression has become a principal strategy for improving the efficiency of Large Visual Language Models (LVLMs). Existing methods primarily define compression according to two criteria: token importance and token similarity.

Early methods emphasize importance-based token pruning, aiming to identify and remove visual tokens that contribute less to model performance. These approaches typically use attention scores or other feature metrics to estimate token importance, as in FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Subsequent work (Shang et al., 2024; Ye et al., 2025) incorporates token similarity to reduce redundancy while preserving critical information, recognizing that important tokens can still be highly similar and therefore repetitive. FrameFusion (Fu et al., 2024b), for instance, combines similarity-based merging with importance-based pruning. VisionZip (Yang et al., 2025) reduces redundancy by selecting dominant tokens and merging contextual information, while VisPruner (Zhang et al., 2025b) leverages visual cues to remove redundant items.

Although these methods substantially advance visual-token redundancy reduction, they do not fully address the spatio-temporal similarity structure of video tokens. Video information is redundant both spatially and temporally, yet similarity-driven methods such as FrameFusion mainly emphasize temporal similarity between adjacent frames and do not fully exploit joint spatio-temporal structure. This limitation constrains their ability to capture inherent video redundancy, particularly under aggressive token-reduction settings. ST-SimDiff addresses this gap by modeling spatial and temporal token relationships within a single graph and by supplementing redundancy compression with explicit preservation of temporal differences.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) encodes the video into a full sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This representation is computationally expensive because self-attention scales quadratically with the token count N , i.e., $O(N^2)$. The task is therefore to construct an efficient token selection function, $f(\cdot)$, that maps the full sequence T_{full} and a compression ratio r to a compact subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The objective is to maximize downstream LVLM performance under the budget constraint that $|T_{\text{sub}}|$ is approximately $r \cdot N$. The central design problem is not merely to remove tokens, but to preserve the information required for video reasoning within the compressed subset T_{sub} .

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, ST-SimDiff consists of two parallel components operating on a spatio-temporal graph constructed from video tokens: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). SRTS compresses stable and redundant content by applying community detection to the graph, grouping highly similar tokens into

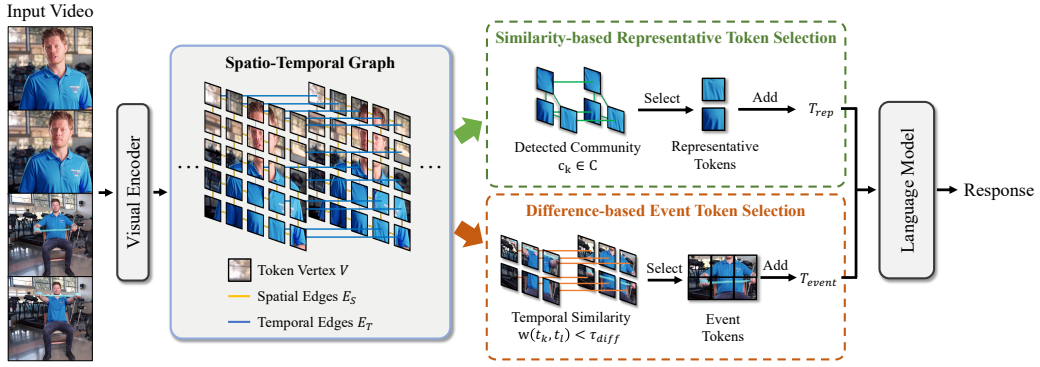


Figure 2: Overview of ST-SimDiff, which constructs a spatio-temporal token graph and performs two coordinated selection operations: Similarity-based Representative Token Selection and Difference-based Event Token Selection.

clusters that correspond to persistent visual elements such as static backgrounds. Within each cluster, highly central tokens are retained to form the representative set T_{rep} . DETS captures dynamic shifts by analyzing temporal graph edges. It identifies locations where the similarity between corresponding tokens in adjacent frames drops sharply; these high-difference points indicate temporal turning points, and the tokens associated with them are retained as event evidence in T_{trans} . The final subset $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$ combines representative stable content with event-defining transitions, yielding a compact visual input that preserves both persistent context and critical temporal change.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder produces a token sequence $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ corresponds to a specific spatio-temporal location and has an associated feature vector $\mathbf{x}_k$. We represent the location of t_k by its frame index $T(t_k)$, spatial height index $H(t_k)$, and spatial width index $W(t_k)$.

We encode token relationships with a spatio-temporal graph $G = (V, E)$, where the vertex set V contains all tokens in T . The edge set E is the union of two subsets, $E = E_S \cup E_T$, which encode spatial adjacency and temporal continuity, respectively. The spatial edge set E_S connects tokens that are adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edge set E_T connects tokens at the same spatial position in adjacent frames:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

Each edge weight $w(v_i, v_j) \in E$ is defined as the cosine similarity between the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph simultaneously encodes local spatial structure and temporal continuity, providing the unified substrate on which the two selection paths operate.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content contains substantial spatio-temporal redundancy. In a static scene, background tokens remain highly similar across time; for a moving object, semantically corresponding tokens may remain correlated even as their spatial positions change. Existing methods often treat temporal and spatial similarities separately, whereas the proposed graph represents their joint structure. Within

this graph, semantically related tokens form dense “communities” or “clusters” according to feature similarity, regardless of their specific frame or location. Selecting central representatives from each community therefore provides an effective mechanism for compressing redundant information while retaining the semantic content of stable visual regions.

To identify strongly correlated token clusters, we first threshold the graph G . We set a similarity threshold τ_{sim} and retain only edges whose weights exceed this threshold, obtaining $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. We then apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) to G' to obtain token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, tokens are ranked by centrality. The centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is its average similarity to all other tokens in the same community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

The intra-community retention rate is set to the global compression ratio r , and the top $\lceil |c_k| \cdot r \rceil$ tokens with the highest centrality scores are preserved from each community c_k . This uniform filtering process also handles single-node communities composed of unique tokens, for which the sole node is retained. Formally, the representative-token set T_{rep} is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the set of k elements with the highest scores from a given set. This strategy compresses static and persistent video content by retaining the most central tokens within each semantic cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity characterizes the stable content of a video, difference characterizes its events. A compression strategy based only on similarity can preserve “what is” while weakening evidence about “what happened”. Key video events are defined by change, including the appearance of a new object, the start of an action, or a scene transition. In the proposed graph, such changes appear as sharp feature differences between temporally adjacent tokens. Capturing these temporal turning points is therefore essential for preserving dynamic evidence relevant to questions such as “when” and “why”.

Temporal difference often marks the occurrence of a key event. The entry of a new object, a scene change, or the beginning or end of an action can substantially alter visual features at corresponding positions in adjacent frames. We therefore analyze the temporal edges (E_T) in the spatio-temporal graph. For any temporal edge $(v_k, v_l) \in E_T$ connecting adjacent tokens t_k and t_l , we apply a difference threshold τ (e.g., the 95th percentile of all temporal edge difference scores). When the temporal similarity falls below the difference threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$, the subsequent token t_l is retained as a critical event token. Formally, the event-token set T_{event} is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This selection rule preserves tokens that mark substantial temporal change, ensuring that dynamic aspects of the video are retained rather than being suppressed by redundancy-oriented compression.

4.5 OVERALL REDUCTION PROCESS

The framework first computes the representative token set T_{rep} and the event token set T_{event} in parallel, and then forms the initial candidate set as their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$. To satisfy the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$, we apply a final pruning step. If $|T_{\text{candidate}}|$ exceeds N_{target} , the method removes the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens with the lowest importance. Following existing work (Chen et al., 2024), token importance is determined by its attention score in the shallow layers of the LLM. This step allows ST-SimDiff to meet a specified computational budget while preserving the tokens selected by the graph-based similarity and difference criteria as much as possible.

Overall, ST-SimDiff combines graph-based token selection with attention-based budget adjustment. The graph stage uses the intrinsic structure of video content, expressed through similarity and difference, to retain representative and event-critical information; the pruning stage provides a flexible mechanism for matching the requested token budget. This design makes the compression process both principled and adaptive.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of ST-SimDiff is analyzed across three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

Graph construction computes cosine similarity between each of the N visual tokens and its spatially and temporally adjacent neighbors. Because each token has a small, constant number of neighbors, this stage requires a single pass over the token sequence and has complexity $O(Nd)$, where d denotes the feature dimension.

After graph construction, SRTS identifies token clusters. As detailed in our implementation (Section 5.2), we use a connected components algorithm with near-linear complexity $O(N + |E|)$. Since the number of edges $|E|$ is at most $3N$, this reduces to $O(N)$. The subsequent intra-community filtering step can be more expensive if implemented naively, because computing all-pairs similarity within each community c_k has complexity $O(|C| \cdot |c_k|^2 \cdot d)$. To avoid this cost, we constrain the maximum community size. In the rare case that a community exceeds a predefined threshold (e.g., $|c_k| > \sqrt{N}$), it is partitioned so that filtering remains efficient and does not dominate the overall procedure. Although graph clustering algorithms such as Louvain and Leiden provide more elaborate community definitions, their computational complexity is higher, approaching $O(N \log N)$. We therefore use connected components to minimize preprocessing overhead.

DETS performs a single pass over all temporal edges and identifies significant changes by comparing temporal token similarities against the threshold. This process has linear complexity $O(Nd)$.

Consequently, the total complexity of ST-SimDiff is governed by linear-time operations and is $O(Nd)$. This cost is substantially lower than the quadratic self-attention complexity, $O(N^2d)$, incurred by the Large Language Model. ST-SimDiff therefore provides substantial token reduction with limited additional computation relative to the main inference cost.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To evaluate the effectiveness of ST-SimDiff across model backbones and compression regimes, we use LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as base models and compare against representative video-token compression methods. The baselines cover **importance-based reduction methods**, including FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods**, including VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). We also report the original uncompressed model (Vanilla) as the full-token reference and Random Sampling with the same token budget as a lower-bound compression strategy.

Benchmarks We evaluate long-video understanding on three challenging benchmarks: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME covers diverse video types and reports performance over Short, Medium, and Long length categories. LongVideoBench evaluates retrieval and reasoning over long videos through a “referring reasoning” task, with videos of four lengths: 15s, 60s, 600s, and 3600s. EgoSchema is a long-term first-person video question-answering dataset that tests understanding of character intentions and causal chains of actions.

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison on long-form video understanding benchmarks using LLaVA-Video-7B with 64 input frames (%). The best result among compression methods is emphasized in **bold**.

5.2 IMPLEMENTATION DETAILS

Experiments are conducted on NVIDIA L20 GPUs with 48GB VRAM under Ubuntu 22.04. Inference evaluation uses the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video, we follow the original setting of 64 input frames; for NVILA, we also use 64 frames to keep computational-efficiency and resource-usage comparisons consistent. Unless otherwise specified, the similarity threshold is $\tau_{\text{sim}} = 0.8$ and the difference threshold is $\tau_{\text{diff}} = 0.2$. We evaluate token retain ratios of $r = 30\%$ and $r = 50\%$. In the practical implementation, we use connected component finding as the community detection algorithm to balance runtime overhead and performance.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

We compare ST-SimDiff with mainstream efficient video language model (VLM) compression methods on VideoMME, LongVideoBench, and EgoSchema. To assess whether the proposed selection principle transfers across model architectures, we report results on LLaVA-Video-7B and NVILA-8B under token retain ratios of 30% and 50%. Table 1 presents the LLaVA-Video-7B results, and Table 2 presents the NVILA-8B results.

(1) ST-SimDiff delivers the strongest overall results across the reported compression settings. On LLaVA-Video-7B, ST-SimDiff achieves 63.2 on VideoMME, 57.5 on LongVideoBench, and 56.0 on EgoSchema at a 30% token retain ratio, and further reaches 63.8, 57.9, and 57.3 at a 50% retain ratio. On NVILA-8B, it achieves 60.2, 55.2, and 51.7 at 30%, and 61.7, 56.5, and 52.5 at 50%. These results show that the method remains effective across both base models and both compression budgets.

(2) The comparison also clarifies the limitations of existing reduction criteria. Among importance-based methods such as FastV and FasterVLM, FasterVLM is usually stronger, but these approaches can retain tokens that are important yet temporally repetitive, limiting their compression efficiency for video. Hybrid methods such as PruMerge, VisionZip, and FrameFusion are more competitive,

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison on long-form video understanding benchmarks using NVILA-Video-8B with 64 input frames (%). The best result among compression methods is emphasized in **bold**.

with FrameFusion generally providing the strongest baseline performance. However, these methods still primarily optimize for representative “commonality” information and therefore can underemphasize narrative evidence associated with “turning points” and “abrupt events”.

(3) Overall, the baseline results indicate that prior work increasingly captures video “redundancy” but still leaves two issues unresolved: the absence of joint spatio-temporal similarity analysis, which limits modeling of complex redundancy, and the lack of explicit “difference” detection, which weakens preservation of plot-defining changes. ST-SimDiff improves performance by addressing both issues in one framework: the spatio-temporal graph models local spatial and temporal relations jointly, while the difference path actively retains tokens associated with temporal transitions.

5.4 ABLATION STUDY

To validate the effectiveness of each core component within our ST-SimDiff framework, we conducted a series of detailed ablation studies. We started with a Baseline strategy that includes only fundamental importance-based pruning and progressively introduced our proposed similarity selection module (+Sim) and difference selection module (++Diff) to quantify their respective contributions. All experiments were conducted at token retain ratios of 30% and 50%, with the results presented in Table 5.

(1) First, we evaluated the effectiveness of the +Sim module. The experimental results clearly indicate that introducing similarity modeling on top of the baseline consistently improves model performance. We further broke down the similarity strategy into three types: spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.). The data shows that collaboratively modeling the spatio-temporal associations between visual tokens is the most effective way to compress redundancy.

(2) After integrating the optimal spatio-temporal joint similarity module, we introduced the difference selection module (++Diff) to form our complete framework. The addition of this module

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component ablation results (%). The table isolates the effect of similarity modeling (+ Sim) and the additional contribution of difference-based event selection (++ Diff) under 30% and 50% token retain ratios.

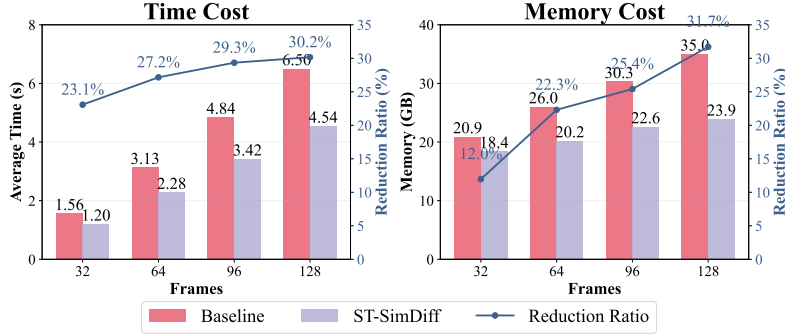


Figure 3: Computational cost comparison between our method and the baseline LLaVA-Video model in average inference time and peak GPU memory usage under a 30% token budget.

provides a significant performance leap, particularly under the high compression ratio of 30%. This demonstrates its vital complementary role: while the aggressive +Sim module focuses on redundancy, the ++Diff module acts as a safety net to identify and preserve unique event tokens that define the video’s narrative. In contrast, the module’s marginal gain is smaller at a 50% retention ratio. This is because the more lenient +Sim module is more likely to have already captured these event tokens, and the overall model performance is already approaching its upper bound. This analysis confirms our core motivation: the framework’s strength lies in its ability to balance similarity-based redundancy compression with the crucial preservation of difference-based events, especially in demanding high-compression scenarios.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify the efficiency advantages of our proposed ST-SimDiff method in practical applications, we conducted detailed tests on its inference time and peak GPU memory usage under a 30% token budget. We compared it against the baseline model (Baseline LLaVA-Video) without any compression. The experimental results in Figure 3 cover video inputs lengths from 32 to 128 frames.

Inference Time ST-SimDiff demonstrates a significant speedup. As the number of input video frames increases, the average inference time of the baseline model grows sharply from 1.56 seconds

(32 frames) to 6.50 seconds (128 frames). In contrast, our method effectively curtails this growth, with an average time of only 4.54 seconds when processing 128 frames. This means that ST-SimDiff achieves an increasingly higher time-saving rate for longer videos, improving from 23.0% for 32 frames to 30.2% for 128 frames.

Memory Cost ST-SimDiff also performs exceptionally well. The peak GPU memory usage of the baseline model climbs linearly with video length, from 20.9 GB (32 frames) to 35.0 GB (128 frames). Our method significantly reduces memory pressure by substantially decreasing the number of tokens fed into the large language model. When processing 128-frame videos, ST-SimDiff’s peak memory usage is only 23.9 GB, saving 31.7% of the memory space compared to the baseline. This advantage enables our model to process longer videos on hardware with limited memory, greatly expanding its potential applications.

In summary, ST-SimDiff not only achieves leading performance but also demonstrates significant advantages in computational efficiency (both time and memory), proving its efficiency for video understanding.

6 CONCLUSION

In this paper, we address the computational inefficiency of Multimodal Large Language Models when processing long videos, a problem rooted in the excessive number of visual tokens and the neglect of critical content changes. We propose ST-SimDiff, a framework requiring no training that builds a spatiotemporal graph to model complex associations between tokens. The method utilizes a dual selection strategy where similarity based selection identifies redundancy via community detection and temporal difference based selection preserves tokens at key transition points. Extensive experiments on VideoMME and LongVideoBench show that ST-SimDiff significantly reduces processing time and memory usage while consistently surpassing state of the art performance. This work introduces a new research direction by balancing content representativeness with transitional events to create a more efficient representation of video data.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

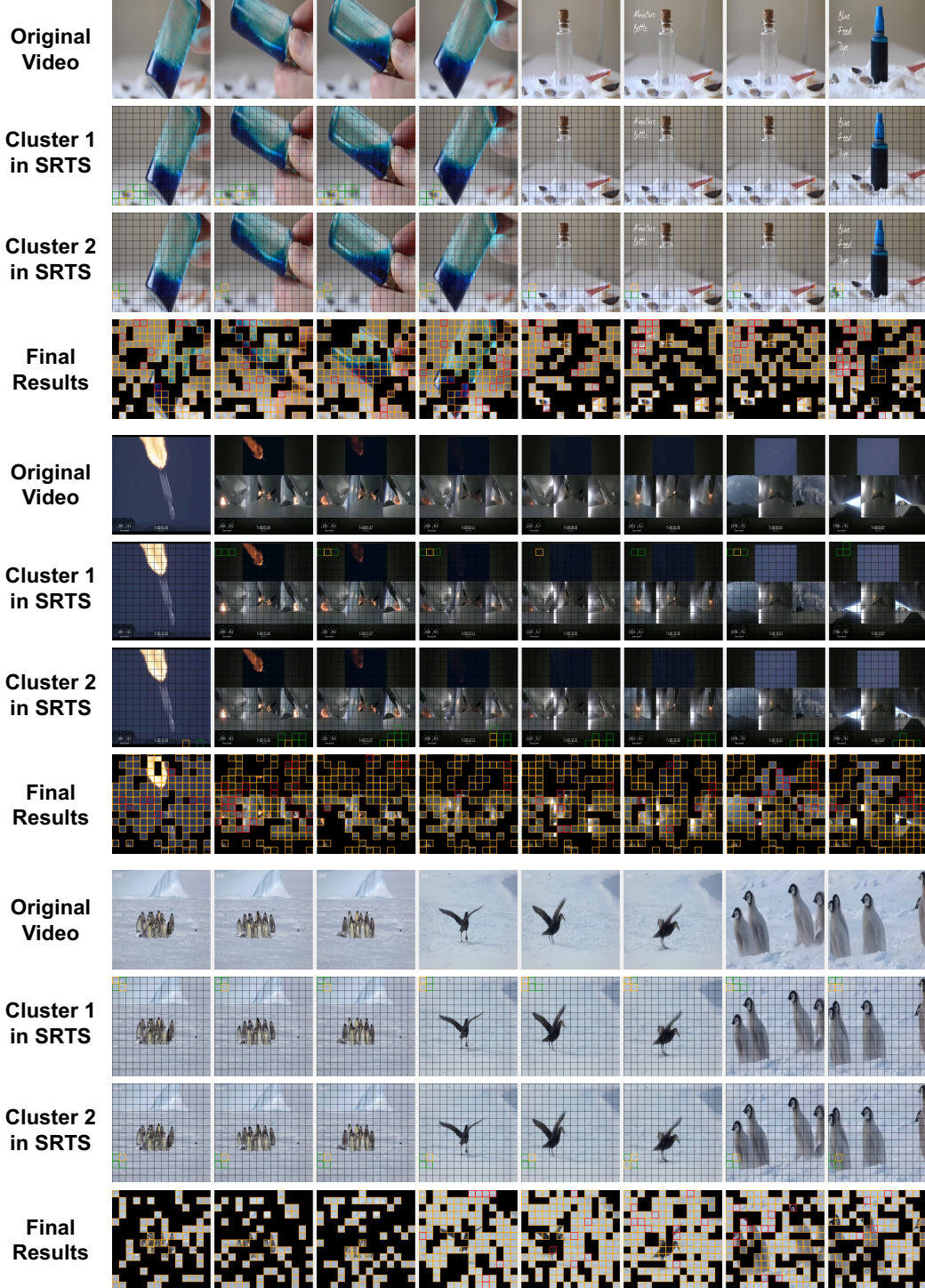


Figure 4: Visualization of the ST-SimDiff process. The visualization breaks down the token selection into communities detected by SRTS (shown as grids in Cluster 1 and 2) and event-driven tokens detected by DETS. The final result highlights the synergy between sparse representative tokens (yellow) for stable content and dense event tokens (red) for dynamic actions.

To provide a more intuitive understanding of the ST-SimDiff framework, we present some visualization samples of the intermediate and final token selection results in Figure 4. The figure illustrates the process on a sample video sequence featuring a dynamic object manipulation task. First, regarding the Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” demonstrate how our graph community detection algorithm functions. It successfully groups spatially and temporally redundant regions, such as the static background and stationary objects, into cohesive communities. This visualization confirms that SRTS effectively identifies stable visual elements and compresses them by selecting only a few central “representative tokens,” which appear as sparse yellow boxes in the final results. This mechanism allows the model to handle massive background redundancy efficiently.

Complementing this, the Difference-based Event Token Selection (DETS) plays a critical role during dynamic moments. As seen in the “Final Results” row, tokens triggered by high temporal difference are explicitly marked with dense red bounding boxes. These red boxes align perfectly with the moving hand and the shifting blue liquid, verifying that our method precisely locates “turning points” where the temporal difference exceeds the threshold (τ_{diff}). This ensures that fine-grained action details and rapid visual changes are preserved, allowing the model to capture the essence of the event without being overwhelmed by static data.

Finally, the synergistic effect of these two components is evident in the overlay of the final results. The visualization demonstrates a powerful balance: the sparse yellow tokens represent the stable background, while the dense red tokens capture the rapid motion. This contrast provides direct visual evidence of our core motivation. ST-SimDiff avoids the blind spots of similarity-only approaches by densely sampling during rapid changes, while simultaneously pruning the spatiotemporal redundancies that importance-based methods often miss. Consequently, our framework achieves an optimal balance between computational efficiency and the preservation of critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Performance comparison on long-form video understanding benchmarks on Qwen2.5-VL-7B for 64 and 128 input frames (%).

To further validate the generalization capability of our proposed framework, we conducted additional experiments on the recently released Qwen2.5-VL model. These evaluations were performed on the VideoMME and LongVideoBench benchmarks to confirm that our method’s efficacy is not limited to a specific model architecture but can be broadly applied.

The empirical results, presented in Table 4, demonstrate the strong performance of ST-SimDiff on the Qwen2.5-VL model. At a token retention ratio of 30%, our method achieved scores of 62.4 on VideoMME and 56.8 on LongVideoBench. When the token budget was increased to a 50% retention ratio, the performance improved to 62.9 on VideoMME and 58.4 on LongVideoBench. It is particularly noteworthy that at the 50% ratio, the performance on VideoMME matches the upper-bound performance achieved using all tokens, and the score on LongVideoBench is highly comparable to its full-performance counterpart. These findings affirm the excellent generalization ability of our framework.

C ABLATION STUDY RESULTS ON NVILA

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	59.3	53.8	50.2
+ Sim	Spatial	59.5	54.5
	Temporal	59.6	54.3
	Spa. + Tem.	59.9	54.6
++ Diff (Proposed)	60.2	55.2	51.7
Token Retain Ratio $r = 50\%$			
Baseline	60.9	55.3	51.5
+ Sim	Spatial	61.1	55.7
	Temporal	61.3	55.6
	Spa. + Tem.	61.4	56.0
++ Diff (Proposed)	61.7	56.5	52.5

Table 5: Ablation results of different components on NVILA-Video-8B model (%).

To further validate the generalization ability of each component in our framework across different base models, we also conducted a series of detailed ablation studies on the NVILA-Video-8B model (Liu et al., 2024c). The experimental setup is consistent with the ablation study in the main paper; we start with a Baseline strategy that includes only fundamental importance-based pruning and progressively introduce the similarity selection module (+ Sim) and the difference selection module (++) Diff).

As shown in Table 5, the experimental results once again validate the effectiveness of our framework design, and the trends are highly consistent with the performance on LLaVA-Video-7B. First, after introducing the similarity module (+ Sim) on top of the baseline, the model’s performance shows a steady improvement at both 30% and 50% token retain ratios. Within the similarity module, we compared three strategies: spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.). The results show that spatio-temporal joint modeling (Spa. + Tem.) is the most effective. For example, with $r = 50\%$, the spatio-temporal joint strategy achieved a score of 61.4% on VideoMME, outperforming the spatial-only and temporal-only strategies. This again demonstrates the importance of collaboratively considering spatio-temporal associations for effectively identifying redundant information. After integrating the optimal spatio-temporal joint similarity module, we further introduced the dissimilarity selection module (++) Diff) to form our complete ST-SimDiff framework. The table clearly shows that the addition of this module brought the most significant performance gains. This result strongly demonstrates the effectiveness and generalization ability of the two core modules in our framework (similarity and dissimilarity selection) and once again validates our core idea of simultaneously handling redundancy and change.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

In our framework, the similarity threshold τ_{sim} and the difference threshold τ_{diff} are two key hyperparameters that respectively influence the selection of representative and transitional tokens. To investigate their impact, we conducted a series of ablation studies, with the results shown in Figure 6. The experiments show that the impact of both parameters on model performance follows a similar trend, first rising and then falling, while demonstrating good robustness within a certain range. For τ_{sim} , a value that is too low leads to imprecise community detection, while a value that is too high can disrupt the integrity of semantic clusters. For τ_{diff} , a value that is too low can introduce noise due to over-sensitivity, whereas a value that is too high may cause the model to miss key events. According to the experimental results, the model achieves optimal performance at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$. Therefore, we adopted this optimal configuration for all other experiments.

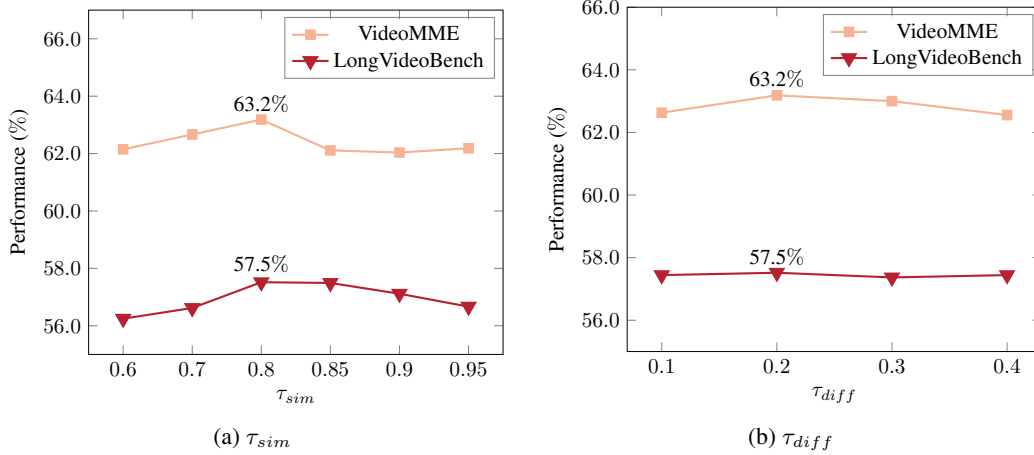


Figure 5: Ablation study results for different values of τ_{diff} and τ_{sim} on VideoMME and LongVideoBench.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

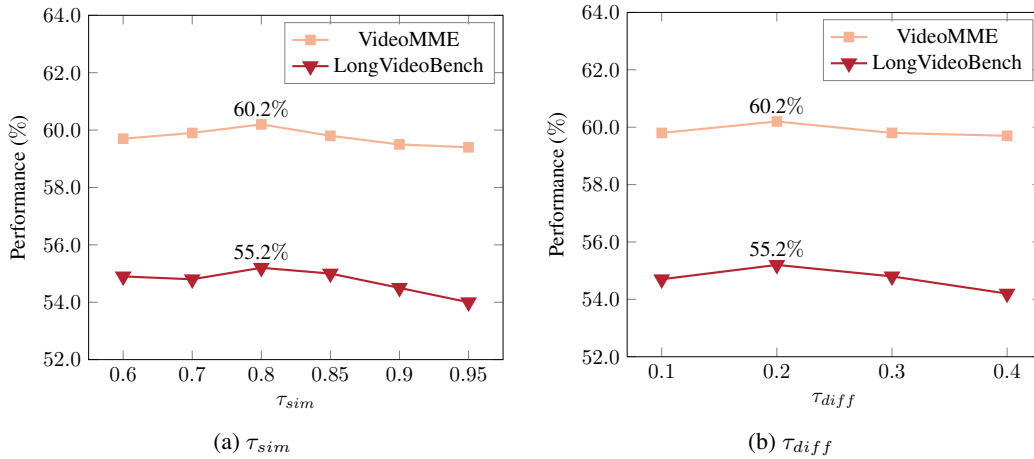


Figure 6: Ablation study results for different values of τ_{diff} and τ_{sim} on NVILA-Video-8B.

To determine the robustness of the key hyperparameters in our method, we conducted a series of ablation studies on the NVILA-Video-8B model. As shown in Fig. 6, we evaluated the impact of the similarity threshold, τ_{sim} , and the difference threshold, τ_{diff} , on the model’s performance across two benchmarks: VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . We tested different values for τ_{sim} within the range of $[0.6, 0.95]$. The experimental results show that the model’s performance is relatively insensitive to changes in this threshold. On the VideoMME dataset, peak performance of 60.2% was achieved at $\tau_{sim} = 0.8$. Similarly, on the LongVideoBench dataset, performance was also optimal at $\tau_{sim} = 0.8$, reaching 55.2%. While performance slightly decreased when the threshold was too high or too low, the overall fluctuation was minimal, confirming that our method maintains good stability across a wide range of similarity thresholds.

Impact of Difference Threshold τ_{diff} . We tested τ_{diff} in the range of $[0.1, 0.4]$. Similar to τ_{sim} , the model’s performance demonstrated strong robustness to variations in τ_{diff} . In both benchmarks, the optimal performance was achieved around $\tau_{diff} = 0.2$, with VideoMME at 60.2% and LongVideoBench at 55.2%. This indicates that setting the difference threshold around 0.2 is most

effective for capturing key events in the video, thereby maximizing model performance while compressing tokens.

In summary, these ablation studies confirm the robustness of our proposed method with respect to its key hyperparameters. The results validate our choice of using $\tau_{\text{sim}} = 0.8$ and $\tau_{\text{diff}} = 0.2$ in our main experiments, as these values consistently yield optimal performance across different benchmarks and different base models.